Credit Card Default Auto-ML Report

🕐 2026-05-28 22:52:29 UTC | 🏆 Best: **catboost** roc_auc = 0.7796 | 학습 24000행

테스트 6000행 | Feature 19개

1 모델 비교 (holdout 기준)

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc	best_iter (avg)
lgbm	0.7791	0.4349	3.1123	0.8168	0.6652	0.3459	0.4551	0.5563	175
xgb	0.7777	0.4265	3.1123	0.8195	0.6723	0.3587	0.4678	0.5548	175
catboost BEST	**0.7796**	**0.4362**	**3.1274**	**0.8177**	**0.6589**	**0.3640**	**0.4689**	**0.5588**	**305**
elasticnet	0.7314	0.3895	2.9314	0.8072	0.6580	0.2668	0.3796	0.4988	-
ensemble	0.7663	0.4168	3.0595	0.8153	0.6585	0.3429	0.4509	0.5377	-

2 CV (OOF) 비교

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc
lgbm	0.7858	0.4343	3.1852	0.8216	0.6786	0.3679	0.4771	0.5584
xgb	0.7851	0.4325	3.2002	0.8217	0.6843	0.3598	0.4716	0.5560
catboost BEST	**0.7841**	**0.4356**	**3.1946**	**0.8220**	**0.6769**	**0.3733**	**0.4812**	**0.5586**
elasticnet	0.7463	0.4038	2.9591	0.8084	0.6687	0.2654	0.3800	0.5075
ensemble	0.7753	0.4245	3.1155	0.8190	0.6804	0.3428	0.4559	0.5465

3 오버핏 점검 (Train vs Holdout)

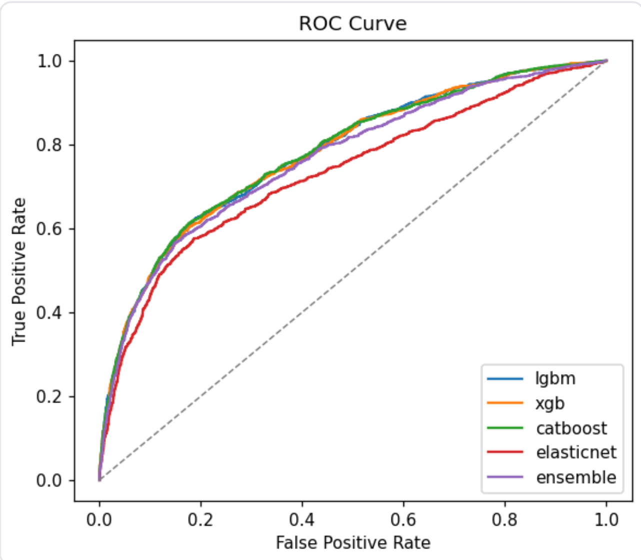
metric	lgbm			xgb			catboost BEST				
	Train	Test	Δ	Train	Test	Δ	Train	Test	Δ	Train	Te
roc_auc	0.8144	0.7791	+0.0353	0.8222	0.7777	+0.0445	0.8099	0.7796	+0.0303	0.7482	0.731
ks	0.4739	0.4349	+0.0389	0.4841	0.4265	**+0.0576**	0.4660	0.4362	+0.0297	0.4062	0.389
lift	3.2775	3.1123	**+0.1652**	3.3189	3.1123	**+0.2066**	3.4037	3.1274	**+0.2763**	2.9780	2.931
accuracy	0.8252	0.8168	+0.0083	0.8278	0.8195	+0.0083	0.8308	0.8177	+0.0131	0.8086	0.807
precision	0.7053	0.6652	+0.0401	0.7078	0.6723	+0.0355	0.7133	0.6589	**+0.0543**	0.6702	0.658
recall	0.3601	0.3459	+0.0143	0.3773	0.3587	+0.0186	0.3931	0.3640	+0.0291	0.2652	0.266

metric	lgbm			xgb			catboost BEST				
	Train	Test	Δ	Train	Test	Δ	Train	Test	Δ	Train	Te
f1	0.4768	0.4551	+0.0217	0.4922	0.4678	+0.0244	0.5069	0.4689	+0.0379	0.3800	0.379
pr_auc	0.6021	0.5563	+0.0458	0.6174	0.5548	**+0.0626**	0.6194	0.5588	**+0.0606**	0.5102	0.498

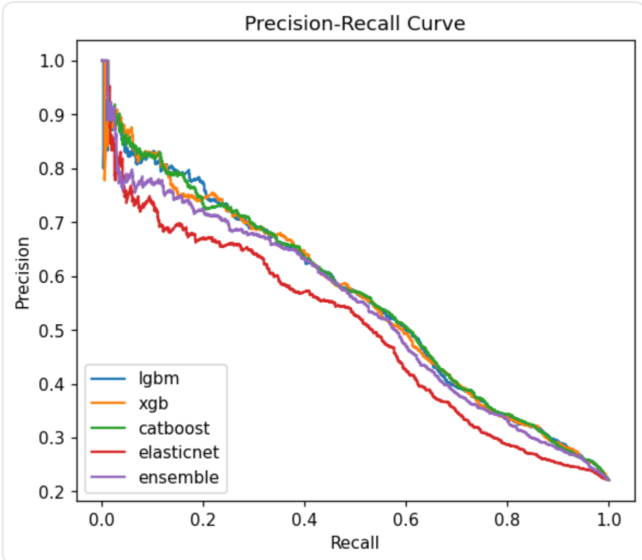
Δ = Train − Test. Train 은 final 모델을 학습 데이터에 fit 한 뒤 동일 학습 데이터로 in-sample 스코어링한 결과로, 의도적으로 over-optimistic 한 추정치다. Test 와의 양의 gap 이 클수록 메모리제이션(overfit) 신호이며, |Δ| ≥ 0.05 / |Δ| ≥ 0.10 인 셀은 색으로 강조한다. (정확한 임계치는 도메인/지표에 따라 다르며 본 색은 참고용.)

4 곡선 비교

ROC Curve



Precision-Recall Curve



5 Best 모델 상세 — catboost

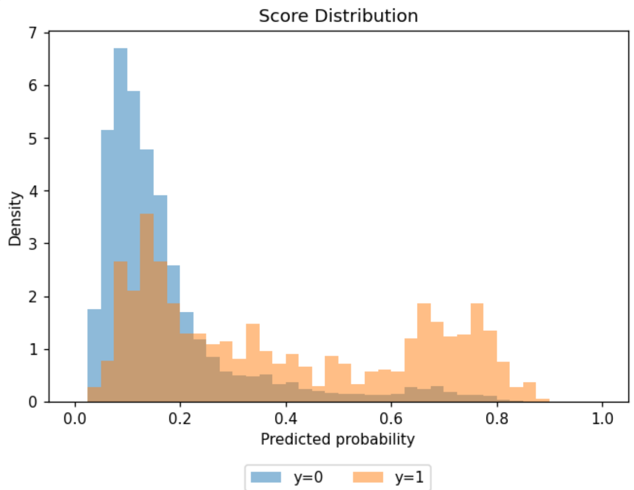
Feature Importance (relative, top 19 · 전체 분포는 feature_importance.csv)

Rank	Feature	Importance
1	PAY_0	0.2307
2	LIMIT_BAL	0.1047
3	BILL_AMT1	0.0909
4	PAY_2	0.0574
5	PAY_3	0.0499
6	PAY_AMT3	0.0471
7	PAY_AMT2	0.0468
8	BILL_AMT2	0.0409
9	PAY_AMT1	0.0408
10	PAY_4	0.0395
11	AGE	0.0342
12	PAY_AMT6	0.0327
13	PAY_6	0.0313
14	PAY_5	0.0298
15	PAY_AMT4	0.0285

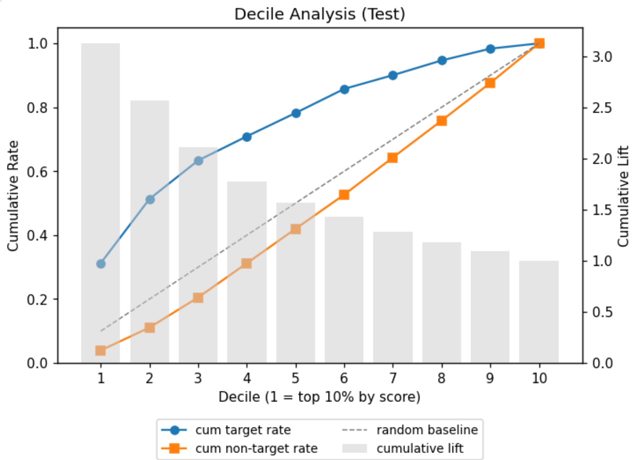
Rank	Feature	Importance
16	BILL_AMT5	0.0259
17	BILL_AMT6	0.0255
18	PAY_AMT5	0.0220
19	SEX	0.0213

10-분위 분석 (테스트셋, 점수 내림차순) 및 점수 분포

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.596 ~ 0.886	600	415	185	415	185	600	10.0%
2	0.324 ~ 0.596	600	266	334	681	519	1200	20.0%
3	0.213 ~ 0.324	600	160	440	841	959	1800	30.0%
4	0.170 ~ 0.213	600	100	500	941	1459	2400	40.0%
5	0.144 ~ 0.169	600	97	503	1038	1962	3000	50.0%
6	0.122 ~ 0.144	600	100	500	1138	2462	3600	60.0%
7	0.103 ~ 0.122	600	57	543	1195	3005	4200	70.0%
8	0.086 ~ 0.103	600	61	539	1256	3544	4800	80.0%
9	0.067 ~ 0.086	600	49	551	1305	4095	5400	90.0%
10	0.029 ~ 0.067	600	22	578	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4423	250
True 1	844	483

6 학습 설정 요약

설정 키	값
target_column	default
categorical_columns	SEX, EDUCATION, MARRIAGE
id_columns	client_id
preprocessing.numeric_null_strategy	median
preprocessing.categorical_null_strategy	most_frequent
preprocessing.outlier_method	percentile
preprocessing.outlier_action	clip
preprocessing.skew_method	signed_log1p
preprocessing.skew_threshold	1.0
preprocessing.scaling_method	standard
training.cv_folds	4
training.early_stopping_rounds	50
training.primary_metric	roc_auc

설정 키	값
training.random_state	42
tuning.enabled	True
tuning.n_trials	15
tuning.cv_folds	3
tuning.timeout	None

7 모델 파라미터

lgbm 튜닝 완료 · 15회 best OOF 0.7850

파라미터	값
objective	binary
metric	auc
verbose	-1
n_estimators	800
learning_rate	0.028180680291847244
num_leaves	26
max_depth	9
min_data_in_leaf	50
feature_fraction	0.6488152939379115
bagging_fraction	0.798070764044508
reg_alpha	2.039373116525212e-08
reg_lambda	1.527156759251193
min_gain_to_split	0.12938999080000846
actual_max_depth	10

xgb 튜닝 완료 · 15회 best OOF 0.7855

파라미터	값
objective	binary:logistic
eval_metric	auc
tree_method	hist
n_estimators	800
learning_rate	0.026000059117302653

파라미터	값
max_depth	7
subsample	0.6563696899899051
colsample_bytree	0.9208787923016158
min_child_weight	1.6709557931179373
reg_alpha	7.620481786158549
reg_lambda	0.08916674715636537
gamma	0.993578407670862

catboost BEST 튜닝 완료 · 15회 best OOF 0.7847

파라미터	값
loss_function	Logloss
eval_metric	AUC
iterations	800
verbose	False
allow_writing_files	False
learning_rate	0.03220579752389847
depth	6
l2_leaf_reg	4.60313968685407
random_strength	0.723236835259315
bagging_temperature	0.6449759616213921

elasticnet 튜닝 완료 · 15회 best OOF 0.7461

파라미터	값
max_iter	1000
C	7.553503645583172
l1_ratio	0.005297120813991926

ensemble 튜닝 미수행

파라미터	값
weights	{'elasticnet': 0.4840579071352653, 'catboost': 0.5159420928647347}

8 변수 선택 (Stability Selection)

base: **lasso** | 부분표본 **80회** | 임계값 **0.60** | 채택 **19 / 23**

표 노출 23 / 전체 23 (전체 분포는 feature_selection.csv)

Feature	Selection Frequency	채택
LIMIT_BAL	100.0%	✓ 채택
AGE	100.0%	✓ 채택
PAY_0	100.0%	✓ 채택
PAY_2	100.0%	✓ 채택
PAY_3	100.0%	✓ 채택
PAY_4	100.0%	✓ 채택
BILL_AMT1	100.0%	✓ 채택
PAY_AMT1	100.0%	✓ 채택
PAY_AMT2	100.0%	✓ 채택
PAY_AMT3	100.0%	✓ 채택
PAY_AMT6	100.0%	✓ 채택
PAY_5	98.8%	✓ 채택
BILL_AMT5	98.8%	✓ 채택
PAY_6	97.5%	✓ 채택
PAY_AMT4	96.2%	✓ 채택
PAY_AMT5	95.0%	✓ 채택
SEX	91.2%	✓ 채택
BILL_AMT6	88.8%	✓ 채택
BILL_AMT2	71.2%	✓ 채택
BILL_AMT3	56.2%	제외
BILL_AMT4	28.7%	제외
EDUCATION	21.2%	제외
MARRIAGE	15.0%	제외